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EXECUTIVE SUMMARY OF THE THESIS

Green Federated Learning: a Data-Centric Approach in the Cloud Continuum

LAUREA MAGISTRALE IN COMPUTER SCIENCE ENGINEERING - INGEGNERIA INFORMATICA

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1. Introduction

The dynamic landscape of Artificial Intelligence (AI) and Machine Learning (ML) has gained widespread attention combined with a world dominated by IoT devices and concepts such as cloud, edge, and Fog Computing. Furthermore, the demand for computational efficiency and sustainability has become paramount. The environmental challenges posed by Deep Learning applications necessitate a shift from Red AI to Green AI [5], emphasizing efficiency in creating a sustainable, data-driven world. Within this context, data quality issues and the need for privacy are dominant. The research project outlined in this thesis aims to address these challenges by exploring the Federated Learning (FL) approach. The thesis focuses on developing a data-centric methodology to optimize data usage, reducing carbon emissions and enhancing FL system efficiency.

2. State of the Art

The AI training cost has increased significantly over the years, as well as the carbon footprint generated making the process environmentally unfriendly. The reaching of state-of-the-art results has left space for research focused on improving the efficiency and reducing the energy

consumption of these models, therefore giving contributions to the Green AI world [5]. DL models require a substantial volume of data for effective training and testing processes. The quality of such data has a profound impact on the precision of DL models [3]. Data quality is defined as the ability of data to meet specific requirements. There are several dimensions to measure the quality depending on a specific task (e.g., accuracy, consistency and completeness) and each of them might contribute differently to the model performance. Recent studies have demonstrated that the size of the dataset used to train Machine Learning models has a significant impact on energy consumption whereas performance is almost unaffected [1]. In [1], these findings have been used to provide a centralized data-centric methodology to reduce carbon emissions of DL. Centralized approaches require the data to be collected in a single location. However, most of the time, data is produced by sensors at the edge of the network. Moving this data is expensive and time-consuming and exposes the data to privacy issues. The need for a decentralized approach has opened the doors to FL, which enables the training of a single DL model across multiple participant nodes. Each node trains a local model and sends its param-

eters to a centralized server responsible for aggregating them into a global model [6]. Not every node involved in a FL configuration always participates. FL poses several challenges, such as the data and hardware heterogeneity of the nodes and the associated expensive communication overhead. The limited computational requirements for training a FL model enable the execution in a Fog Computing [4] infrastructure, a distributed model where each node has distinct performances due to differences in hardware and varying environmental impact based on its geographical location.

This thesis extends the study in [1] to a distributed FL environment. Data reduction is applied by exploiting the heterogeneity in data quality, performance, and energy mix of the different nodes, to reduce the overall environmental impact of the training process.

3. Goals and Requirements

This thesis investigates the environmental impact of FL algorithms in a Fog Computing infrastructure, characterised by high heterogeneity in the hardware (performance, energy consumption, energy mix) and the data (data volume, data quality) of the participating nodes. Two goals have been defined:

- **Goal 1:** identify the data-centric impact on energy consumption and performance in a FL setting with a detailed analysis of the FL model from a data-centric perspective. This includes assessing how variations in data volume influence accuracy and emissions within a FL configuration.
- **Goal 2:** minimize carbon emissions in FL by proposing a methodology for efficient node and data selection, ensuring that the FL algorithm achieves a predefined performance level.

This project also outlines three essential requirements: **R1** - reduce the overall emissions for FL model training; **R2** - satisfy model performance requirements; **R3** - provide a general approach for FL configuration. Goals and requirements drive the development of a robust method for more sustainable and efficient FL systems.

4. Green FL Methodology

The proposed approach aims to create a **FL Configuration Selection System** capable of

providing recommendations and predictions for reducing the environmental impact of FL training. This involves selecting appropriate client nodes and applying data volume reduction. The FL environment consists of N_C heterogeneous nodes located in distinct geographical positions. These nodes own disjoint data subsets and exhibit varying hardware characteristics, resulting in dissimilarities in training performances and energy consumption. The target is a decrease in carbon emissions while either maintaining or improving model accuracy. A researcher can use this approach by providing all the necessary input concerning the dataset and the FL architecture and obtaining an environmentally sustainable FL configuration. The methodology unfolds in three sequential phases: (i) **Data-Centric FL Exploration**; (ii) **Data Analysis and Modeling**; and (iii) **FL Configuration Recommendation** (Fig. 1). The **Data-Centric FL Exploration Phase** exploits a FL simulation environment to extract results regarding system training with distinct datasets and data-centric configurations. The focus is on understanding how data volume and quality impact the FL system's performance and energy consumption during training. All simulation results are reorganized to build curves extracting the relation between data characteristics, final model accuracy and energy consumption. The experiments and the curves are saved for subsequent analysis. This first phase satisfies **Goal 1**. In the **Data Analysis and Modelling Phase**, the obtained results have been analyzed in terms of performance and energy impact. The analysis generates a predictor capable of suggesting how to reduce the size of an unseen dataset provided by a researcher within a Federated setting. This phase is in charge of partially addressing all the goals defined. The subsequent **FL Configuration Recommendation Phase** adapts predictions to the specific execution context defined by the researcher. This phase involves a FL Configuration Recommender, considering the infrastructure's node characteristics and the data subsets they contain. It successfully achieves **Goal 2** by selecting and ranking nodes for efficient FL algorithm training.

An overview of the architecture is shown in Fig. 1. The components of the architecture are split according to the three phases of the ap-

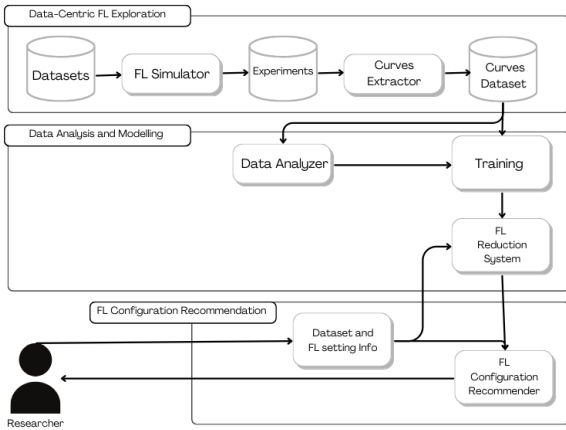


Figure 1: FL Configuration Selection System Architecture.

proach. The *FL Simulator* component is crucial for executing experiments and determining the performance and energy outcomes of FL systems through training with multiple datasets and data-centric configurations. The simulator creates an environment with a central server and participant nodes. These nodes train local models, communicating results to the central server. Experiments have been executed with different datasets. For each of them, sub-experiments have been executed to explore data quality and volume impacts using two different approaches: vertical and horizontal. The horizontal approach investigates a selected data property affecting all the nodes equally, whereas the vertical approach affects only a subset of nodes. Data-related properties are progressively disturbed to observe their impact. Results are stored in the *Experiments* database. The *Curves Extractor* component constructs curves modelling patterns for each experiment, specified by a dataset, experiment type (horizontal or vertical), and data property (e.g., volume, accuracy, consistency, completeness). It builds logarithmic curves according to Eq. 1 through a regression model. An example is shown in Fig. 2. The parameters C_1 and C_2 with the experiment’s characteristics are saved in the *Curves Dataset* database.

$$Curve = C_1 + C_2 \cdot \log(x) \quad (1)$$

The *Data Analyzer* component offers comprehensive insights and explanations concerning empirical observations of energy consumption and accuracy levels in FL model training. It investigates how these metrics relate to pertur-

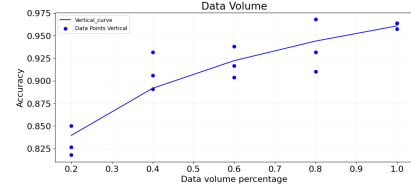


Figure 2: Curve Example.

bations in data volume and data quality dimensions, analyzing the patterns observed in the previously collected experimental data and curves. While determining the effective impact of data properties on the final results of the model, the most convenient approach between horizontal and vertical has been selected. The *Training* component is responsible for building a machine learning regressor to predict data volume reduction for an unseen FL configuration while maintaining a predefined accuracy level. Dataset type, number of training samples, length of the sequence, number of classes, and accuracy are the input features of the model. The target of the model is the data volume necessary to reach the accuracy feature. The resulting regressor model is named *FL Reduction System* component. The researcher submits a FL training task for execution in a distributed infrastructure, along with target accuracy. This component predicts either the percentage of data volume or the number of nodes required to meet the specified accuracy threshold. The *FL Configuration Recommender* component uses the predictions from the *FL Reduction System* and the infrastructure details provided by the researcher to generate a FL configuration recommendation. This recommendation includes node selection and data volume reduction to effectively reduce carbon emissions while maintaining the specified accuracy threshold.

5. Implementation

The proposed approach has been tested in the context of time series classification with the usage of multiple datasets retrieved from the UCR/UEA archive¹. The DL model used to execute the local training is ResNet. The performance of the FL model is evaluated using the accuracy metric. Energy costs are expressed in

¹https://www.cs.ucr.edu/%7Eeamonn/time_series_data_2018/

kWh. Carbon emissions are retrieved from the energy consumption and the carbon intensity of the node location and are measured in kgCO_2 equivalents (kgCO_2e).

5.1. Data-Centric FL Exploration

The data-centric FL Exploration phase involves conducting simulations. The *FL Simulator* has been implemented using the framework Flower [2], which enables the simulation of multiple nodes on a single system, incorporating functions for the transmission and aggregation of training parameters. A fixed FL configuration has been used in this phase which includes 10 nodes. For each experiment, defined by a dataset and a type (horizontal or vertical), multiple sub-experiments are executed by the *FL Simulator*. These sub-experiments differ from one another based on the dataset configuration (e.g., data volume and data quality). Throughout the sub-experiment cycle, data-centric characteristics were varied with a progressive percentage reduction on the dataset. At the end of the experiment, accuracy and energy consumption are registered. Results are used by the *Curves Extractor* to build a logarithmic curve. Each curve represents the behaviour of an experiment, with all the related sub-experiments, concerning a dependent variable as shown in Fig. 2.

5.2. Data Analysis and Modelling

The *Data Analyzer* component is in charge of analyzing the results and the curves obtained. For each data property analyzed, a comparison between horizontal and vertical solutions is given concerning accuracy and energy.

The first comparison analyzed is the variation of data volume related to the accuracy of the model (Fig. 3). The vertical approach yields higher or equal accuracy than the horizontal approach.

The trade-off between data volume and energy is shown in Fig. 4. As before, the vertical approach demonstrates a lower environmental impact, due to the involvement of a reduced number of nodes. In conclusion, the vertical approach is preferable to keep a higher level of accuracy and lower energy consumption.

Experiments have also involved data quality dimensions (i.e., accuracy, consistency and completeness). Results in [3] and [1] suggest that different dimensions have a distinct impact on

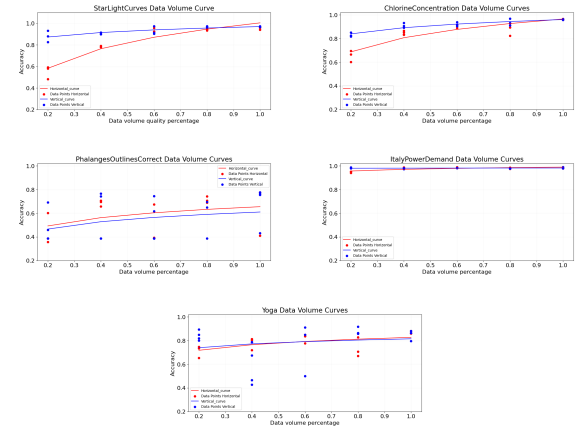


Figure 3: Accuracy and Data Volume trade-off.

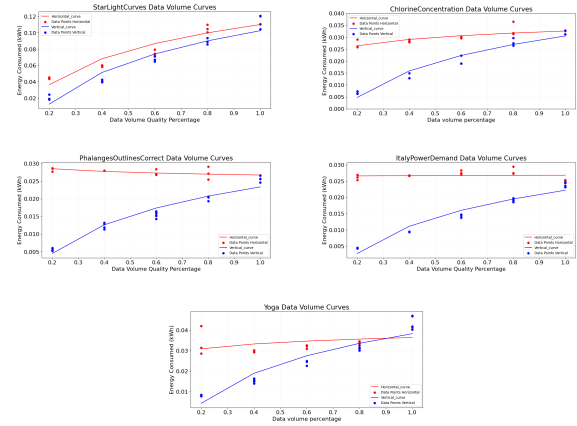


Figure 4: Energy and Data Volume trade-off.

training, with accuracy and consistency more critical than completeness. Experiments confirmed these expectations hold in a FL scenario. Experiments have verified this aspect by conducting comparisons with a single dataset.

The *Training* component builds a regression model that can predict the number of nodes for unseen datasets. Only vertical configurations that demonstrated better performance and lower emissions have been considered. A hyperparameter search approach has been employed to test several machine learning models (Linear Regression Lasso, Ridge, ElasticNet, Random Forest, Decision Tree, Gradient Boosting, and XGB regressor) and the one with the lowest error test has been selected (Gradient Boosting).

5.3. FL Configuration Recommendation

The *FL Configuration Recommendation* phase aims at reducing the carbon footprint while

maintaining a requested performance level for a specific training task submitted by a researcher. The researcher provides as input the FL configuration data, including for each node: power profile, location, dataset volume, and data quality. The component recommends a FL configuration, selecting nodes and portions of the dataset per each node for FL training.

The *FL Configuration Recommender* uses the model generated by the *FL Reduction System* to obtain the prediction of the number $\hat{N}$ of nodes needed to reach the desired performance level. The *FL Reduction System* computes the prediction based on the dataset characteristics presented by the researcher. The prediction is generated considering the homogeneous FL configuration used during training. The number $\hat{N}$ of nodes needs to be adapted to fit the FL configuration indicated by the researcher computing the effective prediction as $N = \frac{\hat{N} \cdot N_C}{10}$. The Target Data Volume percentage V is computed starting from N as $V = N \cdot V_n$, where N_C is the number of nodes in the configuration submitted by the researcher and $V_n = \frac{1}{N_C}$ is the basic volume percentage expected in the single node. The *FL Configuration Recommender* builds a ranking of the nodes provided by the researcher based on a score. Given a single node, the score $Score_n$ assigned to the node is computed as in Eq. 2:

$$Score_n = W_E \cdot (1 - \frac{CO_2}{MaxCO_2}) + \sum_{i=1}^I W_i \cdot Q_i \quad (2)$$

W_E and W_i are weights associated with the carbon emissions of the node and the data quality properties (e.g., consistency and completeness). The sum of the weights must be equal to 1. In this thesis, the weights have been set giving more relevance to the energy contribution.

Once a score is assigned to each node, the component should select the set of nodes to be used in the FL training. Three methods are proposed and compared: **Node Selection (NS)**, **Minimal Smart Reduction (MSR)**, and **Smart Reduction (SR)**. The **NS** method chooses the first N nodes with the highest score. The selected N nodes must satisfy the V_n percentage. If the dataset within a node n belonging to N is larger than V_n requirement, it is randomly reduced equalizing the V_n percentage. Otherwise, if the data volume of n does not satisfy the requirement, n is not selected and makes way for

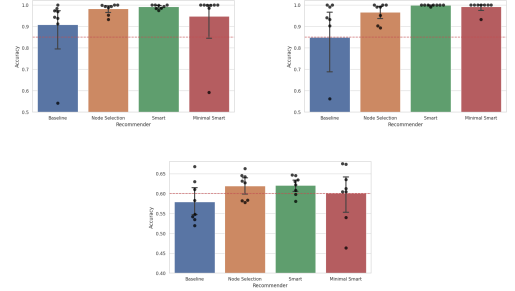


Figure 5: Evaluation Results Accuracy.

the next candidate from the ranking. The **MSR** applies the same strategy, along with the removal of dirty data. Only clean data are used for training. As a consequence, the effective data volume E , which is the sum of the clean data volume of N used for training, might be lower than V , possibly affecting the model accuracy. The **SR** method is similar to the **MSR**, but it satisfies the V target percentage. Removing dirty data from N and summing up the clean data percentage left could result in E be lower than V . The **SR** method selects nodes in excess until E is higher or equal V . The efficiency of the three methods proposed is evaluated in Sec. 6.

6. Evaluation

The evaluation analysis assesses the satisfaction of the goals and requirements presented in Sec. 3. The evaluation needs to be conducted by simulating the role of the researcher in a Fog Computing scenario. Three distinct FL infrastructural configurations with three different datasets have been proposed to test the methods. The three FL configurations consider a distinct number of nodes belonging to the FL setting with different device power, geographical location, data volume and data quality properties. The evaluation compares the results obtained with the three methods proposed in Sec. 5.3 with a Baseline where the FL training is executed with all the nodes and data. The validation experiments have been run and repeated 8 times per each method applied. Considering 3 FL Configurations and applying the 3 methods proposed and the Baseline method for each configuration, a total number of 96 simulations were executed in the FL environment in the evaluation.

According to **Goal 1**, the recommender sys-

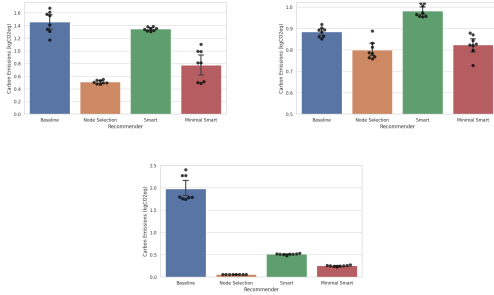


Figure 6: Evaluation Results Carbon Emissions.

tems should satisfy the accuracy threshold set by the researcher. Fig. 5 shows the results of the experiments. The **NS** and the **MSR** methods have satisfied the performance requirement 87% of times whereas the **SR** 92%. An average improvement in the accuracy obtained with the proposed approach compared with the Baseline can be observed. On average, the **NS** method improved the accuracy of 10%, **MSR** of 8% and **SR** of 12%.

Concerning **Goal 2**, emissions of training with the three methods have been compared with the Baseline. Results reveal a good trend for the **NS** and the **MSR** methods, leading to a reduction in emissions 96% of times for the former and 88% for the latter. The **SR** method has reduced the carbon emissions only 33% of the times, producing higher emissions than the Baseline in one experiment. On average the **NS** has yielded a reduction of the carbon emission of 56%, consistently generating less CO_2 than the Baseline, with a peak of 90% reduction. The **MSR** obtains 45% of carbon reduction always producing less CO_2 than the Baseline. On average, the **SR** results in 25% of carbon reduction, violating **Requirement 2** in Configuration 2. **NS** and **MSR** methods demonstrated the ability to minimize the generated carbon emissions while maintaining a high level of performance. The reported data includes the overhead generated by all the steps related to the inference of the *FL Selection Configuration System*.

7. Conclusion

By analyzing data-centric attributes related to energy usage and the achieved model performance in FL training, this thesis proposes a methodology designed to effectively reduce the environmental impact of FL training in hetero-

geneous infrastructures. Starting from a training task and a FL infrastructure provided by a researcher, the proposed approach suggests a FL configuration, with a decreased number of nodes and data employed considering data quality aspects, capable of reducing carbon emissions while ensuring the required level of performance for the final model. Experiments and evaluations have demonstrated that the utilization of this approach can result in a 90% reduction in carbon emissions compared to the Baseline configuration, while maintaining and, most of the time, increasing the accuracy of the trained model. Future work will focus on optimizing the proposed solution. This involves exploring an automatic approach to set weights for selecting nodes in the infrastructure based on their environmental profile and data quality. Furthermore, additional data quality dimensions could be studied to assess their impact on FL and enhance the reduction of data volume.

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